

PinPoint: A Community Driven Map for Local Discoveries

Project Proposal for UCS503
Software Engineering
Thapar Institute of Engineering and Technology

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1 Higher Order Goal

...primary motivation

The project aims to provide a structured way to preserve and discover local geographic knowledge that is often informal, temporary, personal, or difficult to represent using conventional digital maps. The immediate engineering objective is to build a deployable web platform where users can capture, organize, discover, verify, and maintain such locations using geo tagged photographs.

The project is designed as a four month software engineering effort with emphasis on secure architecture, geospatial data management, community workflows, maintainability, testing, and deployment.

2 Time to Value

...development approach

- The project will be developed incrementally, beginning with authentication, maps, geo tagged locations, photographs, and public and private visibility.
- Moderate features such as search, filtering, community verification, conflict handling, and freshness tracking will follow the basic system.
- Advanced AI assisted classification, similarity detection, moderation, and production monitoring will be implemented after the core workflow is stable.
- Git based version control, automated testing, and CI/CD will be integrated throughout development rather than added at the end.

3 Problem Statement

Conventional digital maps are effective at representing established businesses, roads, addresses, and recognized points of interest. However, many useful locations are informal, temporary, locally known, or difficult to represent as conventional places.

Examples include:

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- Underrated food stalls and local vendors
- Scenic viewpoints and sunset locations
- Trail entrances and local shortcuts
- Temporary markets
- Public facilities
- Community gathering spots
- Lesser known local attractions

Information about these locations is commonly shared through personal recommendations, social media, or messaging groups. Such information is therefore difficult to discover, organize, verify, and preserve.

The problem is that useful local geographic knowledge exists, but there is no structured community driven system specifically designed for capturing and discovering these locations while also allowing users to maintain private personal maps.

4 Proposed Solution

...Software Proposition

4.1 Overview

Build a **web based** crowdsourced mapping platform called **PinPoint** with the following components:

- **Public Maps:** users can publish useful local locations for discovery by the wider community.
- **My Maps:** users can maintain private maps containing personal locations, memories, or places that should not be publicly visible.
- **Geo tagged submissions:** users can upload photographs with geographic coordinates and descriptions.
- **AI assisted public classification:** public submissions are assigned tags from the approved AI generated tag suggestions.
- **Community verification:** users can add photographs, confirm whether locations are still available, and report incorrect information.
- **Conflict handling:** nearby or semantically similar locations will be identified to reduce duplicate entries.
- **Admin and Moderator dashboard:** administrators can review reports, flagged submissions, conflicts, and moderation activity.

4.2 Core Workflow

1. User signs in to the platform.
2. User chooses to browse Public Maps or manage My Maps.
3. User creates a location by uploading a photograph, capturing its geographic coordinates, and entering a description.

4. User chooses whether the location belongs to a Public Map or a private My Map.
5. For Public Maps, AI recommends tags from the approved public taxonomy. The user can select only from these suggested tags.
6. For My Maps, the user can define custom tags and descriptions.
7. The system checks nearby and similar existing public locations before publishing.
8. Public locations can subsequently receive additional photographs, verification updates, reports, and freshness information.

4.3 Public and Private Maps

- **Public Maps:** visible to the PinPoint community and subject to AI assisted tagging, moderation, duplicate detection, and community reporting.
- **My Maps:** private by default and accessible only to the owner or users explicitly granted access.
- Private maps can be shared with selected users through explicit access permissions.
- Private locations will not appear in public map searches or public APIs.

4.4 Conflict and Duplicate Handling

When a user attempts to add a public location near an existing location, the system will identify possible conflicts rather than automatically creating a duplicate.

- Display nearby or semantically similar existing locations.
- Allow the user to identify whether the location is the same.
- If it is the same, the user can add another photograph or additional information to the existing location.
- If it is different, the user can continue creating a separate location.

5 Solution Approach

...Engineering focus

This is an engineering focused software project. The primary emphasis will be on secure application architecture, geospatial data management, community workflows, maintainability, and deployability rather than developing novel AI algorithms.

5.1 System Architecture

The system will use a layered web architecture:

- **Frontend:** React with TypeScript and Tailwind CSS.
- **Backend:** FastAPI with Python.
- **Database:** PostgreSQL with PostGIS and pgvector.
- **Object Storage:** AWS S3 for photographs.
- **Caching and Background Processing:** Redis and Celery.

- **Reverse Proxy:** Nginx.
- **Deployment:** Vercel for the frontend and AWS EC2 with Docker Compose for backend services.

5.2 Geospatial Management

Leaflet and OpenStreetMap will provide interactive map functionality. Nominatim will be used for geocoding and reverse geocoding.

PostGIS will store geographic coordinates and support proximity queries used for location discovery and duplicate and conflict detection.

5.3 AI Assisted Public Maps

AI will be used as an assistance and moderation layer.

- **Gemini API:** recommends public categories and tags from user provided descriptions.
- **Sentence Transformers:** generate semantic embeddings for location descriptions and related information.
- **pgvector:** performs similarity searches for potentially duplicate or conflicting locations.
- AI assisted moderation will identify potentially inappropriate, spam, or suspicious public submissions.
- AI will assist moderation but will not override application rules or administrator decisions.

5.4 Authentication and Security

Authentication will be mandatory for all users.

- JWT and OAuth2 for authentication.
- Argon2id for password hashing.
- Ownership and authorization checks for private maps.
- Explicit sharing permissions for selected users.
- Input validation through Pydantic.
- API rate limiting.
- Secure access to stored photographs through controlled URLs.
- HTTPS and Nginx for secure communication.

5.5 Testing and Validation

The system will be validated through multiple levels of testing:

- **Unit Testing:** individual backend functions and services using Pytest.
- **Integration Testing:** API, database, authentication, storage, and background service interactions.
- **End to End Testing:** complete user workflows using Playwright.

- **Security Testing:** authentication, authorization, input validation, rate limiting, and dependency and container security checks.
- **User Acceptance Testing:** testing public and private map workflows, location creation, discovery, verification, and reporting with representative users.

5.6 CI/CD and Development Workflow

The project will use Git and GitHub for collaborative development.

- Feature branches will be used for individual features.
- Pull requests will be used for code review and merging.
- GitHub Actions will automatically run linting, type checking, tests, security checks, and Docker builds.
- Docker and Docker Compose will provide reproducible development and deployment environments.
- Releases will use version tags and semantic versioning.

6 Evaluation Criterion

...measurable and attributable

The system will be evaluated using metrics connected to the primary workflows.

6.1 Primary Evaluation Metrics

- **Location creation success rate:** percentage of valid submissions successfully stored and displayed.
- **Public and private authorization correctness:** percentage of tested private locations that remain inaccessible through unauthorized users and public APIs.
- **Duplicate detection effectiveness:** percentage of known duplicate or nearby conflicting locations correctly surfaced for review.

6.2 Secondary Evaluation Metrics

- API response time for common map and search operations.
- Successful image upload and retrieval rate.
- Community verification and reporting response rate.
- Freshness of public location information.
- Application availability during the deployment period.
- End to end workflow success rate.

7 Scalability

...with foresight

The architecture will support growth in users, locations, photographs, and concurrent requests.

1. Application scalability:

- Separate frontend and backend services.
- Stateless FastAPI APIs.
- Redis based caching.
- Celery based asynchronous processing.
- Pagination and indexed database queries.
- Background execution of expensive AI operations.

2. Infrastructure scalability:

- Managed PostgreSQL through AWS RDS.
- AWS S3 for scalable photograph storage.
- CloudFront for image delivery and caching.
- Docker based deployment.
- CI/CD for repeatable releases.

8 Engine Availability Heuristic

The project uses mature and widely adopted technologies:

- React and TypeScript with Tailwind CSS for frontend development.
- FastAPI and Python for backend APIs.
- PostgreSQL with PostGIS and pgvector for relational, geospatial, and vector data.
- Leaflet and OpenStreetMap for mapping.
- Nominatim for geocoding.
- AWS S3 and CloudFront for image storage and delivery.
- Redis and Celery for caching and background jobs.
- Gemini API and Sentence Transformers for AI functionality.
- Docker and Docker Compose for containerization.
- Pytest and Playwright for automated testing.
- GitHub Actions for CI/CD.
- Sentry and AWS CloudWatch for monitoring.

These technologies allow the team to focus on the application's core workflow, security, geospatial functionality, and community features rather than building infrastructure from scratch.

9 Project Scope and Deliverables

...four month implementation

9.1 Initial Deliverable: Basic

- Project setup and Git based development workflow.
- Mandatory user registration and authentication.
- React based map interface.
- Leaflet and OpenStreetMap integration.
- Creation and viewing of geo tagged locations.
- Photograph upload to AWS S3.
- Basic public and private location visibility.
- Creation and management of My Maps.
- PostgreSQL and PostGIS integration.

9.2 Subsequent Deliverable: Moderate

- Public categories and filtering.
- Location search and detail cards.
- Multiple photographs for locations.
- Nearby location and duplicate suggestions.
- Community verification and reporting.
- Location freshness and status tracking.
- Private map sharing with selected users.
- Admin and Moderator dashboard.
- Redis and Celery background processing.

9.3 Advanced Deliverable

- Gemini based public tag recommendation.
- Sentence Transformer embeddings with pgvector.
- AI assisted duplicate and similarity detection.
- AI assisted spam and content moderation.
- Advanced conflict and freshness handling.
- Security hardening and automated security checks.
- Production CI/CD and monitoring.

9.4 Four Month Timeline

Period	Development Phase	Deliverable
Weeks 1 to 4	Requirements, architecture, database design, project setup, authentication	SRS, architecture, repository, basic authentication
Weeks 5 to 8	Maps, geolocation, image upload, public and private maps	Basic working PinPoint MVP
Weeks 9 to 12	Search, filters, conflicts, community verification, freshness, admin dashboard	Moderate feature complete system
Weeks 13 to 15	AI tagging, embeddings, duplicate detection, moderation	Advanced AI assisted functionality
Week 16	Testing, security checks, deployment, monitoring, documentation	Final release, testing report, presentation

10 Risks and Mitigations

- **Privacy risk:** enforce strict ownership and access control rules for private maps and photographs.
- **Incorrect AI classification:** public tags will be selected only from approved AI suggested tags and remain subject to moderation.
- **Duplicate locations:** combine geographic proximity and semantic similarity to identify possible duplicates for user review.
- **Outdated information:** maintain verification timestamps, status updates, and community reporting.
- **Spam and inappropriate content:** combine AI assisted moderation, rate limiting, reporting, and administrator review.
- **Project scope:** implement features in Basic, Moderate, and Advanced stages so that the core system is completed before advanced functionality.
- **Deployment issues:** use Docker, automated tests, CI/CD, Sentry, and CloudWatch to make deployment repeatable and observable.

11 Summary

PinPoint proposes a deployable web platform for capturing and discovering local geographic knowledge that is difficult to represent through conventional maps. The system combines geo tagged photographs, public community maps, private personal maps, geospatial search, community verification, freshness tracking, and AI assisted public classification and moderation.

The project emphasizes software engineering principles through modular architecture, secure authentication and authorization, automated testing, containerization, CI/CD, monitoring, and managed cloud infrastructure. The four month implementation will progress from a basic mapping system to moderate community features and finally advanced AI assisted functionality.